

Quiz (Habanero)

- Q1.** What is the primary purpose of a stem-and-leaf plot in analyzing temperature changes?
(A) To display the mean temperature
(B) To show the distribution of temperatures
(C) To compare temperatures between cities
(D) To predict future temperature changes
(E) To calculate the standard deviation
- Q2.** A stem-and-leaf plot shows the daily rainfall in a city. If the stem is 20 and the leaves are 5, 7, 2, what are the actual rainfall values?
(A) 25, 27, 22
(B) 205, 207, 202
(C) 20.5, 20.7, 20.2
(D) 2.05, 2.07, 2.02
(E) 2005, 2007, 2002
- Q3.** Which of the following distributions is best represented by a stem-and-leaf plot?
(A) Normal distribution
(B) Skewed distribution
(C) Bimodal distribution
(D) Uniform distribution
(E) Exponential distribution
- Q4.** The stem-and-leaf plot below shows the temperature changes in a region. What is the median temperature?

$$\begin{array}{c|c} 1 & 2, 5, 8, 2 \\ \hline 1, 4, 7 & \end{array}$$
- Q5.** 15^{circ}
Q6. 18^{circ}
Q7. 22^{circ}
Q8. 25^{circ}
Q9. 28^{circ}
- Q10.** A researcher collected data on the amount of CO_2 in the atmosphere. The stem-and-leaf plot shows the following data:

$$\begin{array}{c|c} 3 & 4, 9, 1, 4 \\ \hline 2, 6, 8 & \end{array}$$
 What is the range of CO_2 levels?
- Q11.** $30 - 49$
Q12. $34 - 49$
Q13. $31 - 48$
Q14. $39 - 42$
Q15. $14 - 18$
- Q16.** The stem-and-leaf plot shows the pH levels of different ecosystems. Which ecosystem has the highest pH level?

$$\begin{array}{c|c} 6 & 2, 5, 8, 7 \\ \hline 1, 4, 7 & \end{array}$$
- Q17.** Ecosystem with pH 6.2
Q18. Ecosystem with pH 6.5
Q19. Ecosystem with pH 6.8
Q20. Ecosystem with pH 7.1
Q21. Ecosystem with pH 7.7
- Q22.** Which of the following is a limitation of using a stem-and-leaf plot?
(A) It can only be used for small datasets
(B) It can only be used for large datasets
(C) It does not show the distribution of the data
(D) It is difficult to compare multiple datasets
(E) It is only used for temperature changes
- Q23.** What is the advantage of using a stem-and-leaf plot over a histogram?

Open Ended Questions (FRQ)

- Q9.** Create a stem-and-leaf plot to display the following rainfall data: 23, 27, 19, 31, 22, 28, 25. What can you infer about the distribution of rainfall from the plot?

1, 4, 7. Calculate the mean and median temperature.

- Q10.** The stem-and-leaf plot shows the temperature changes in a region over a month.

Chef's Tips

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- Tip 1: Remind students to check their work carefully
- Tip 2: Encourage students to use the stem-and-leaf plot to identify patterns and trends
- Tip 3: Suggest that students use a calculator to check their calculations